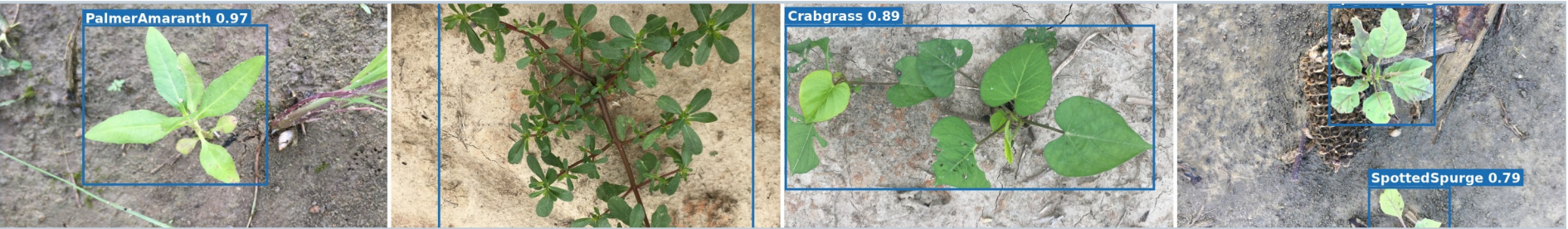


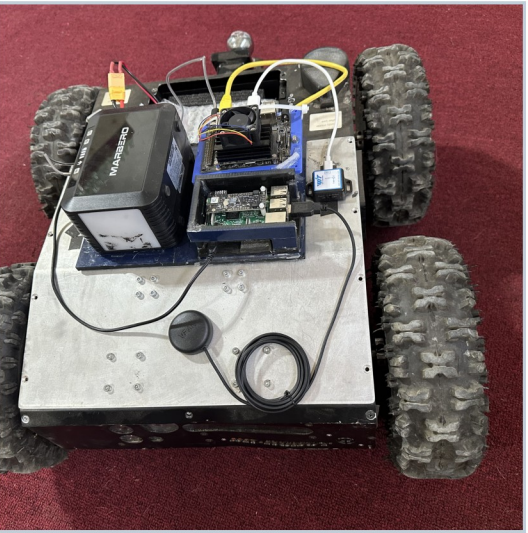
On one platform, agents collect their own data, review one another's work and train without an operator. The loop is built to run without a fixed end; the stop we are working towards is a reviewer judging that nothing further is to be gained. This sheet reports how the agents coordinate and what the collection produced.

Weed Detection



The corpus exceeds 150,000 labelled images. Two agents operate on it unprompted: one harvests and audits imagery, the other trains a detector small enough for the robots.

Rover



A four-wheel skid-steer the width of a crop row carries a Jetson, a Raspberry Pi and its own battery. Camera, GPS and two IMUs stream live over one API key.

Laser Weed Control Robot



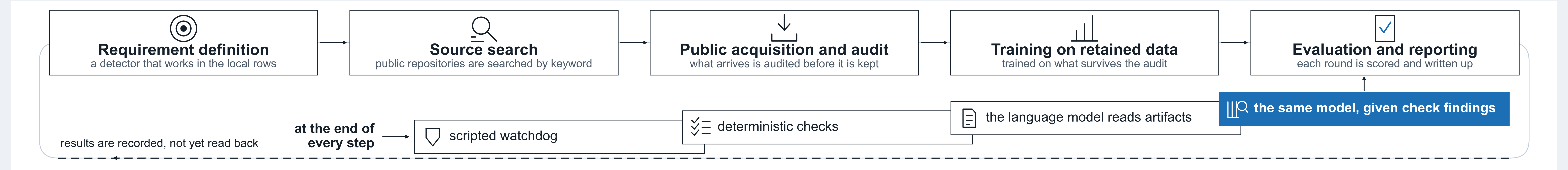
A YOLO detector localises each weed; two galvanometer lasers aim and fire, destroying it in place. No chemical is applied and nothing contacts the adjacent crop.

Humanoid and Quadruped



This project pairs a humanoid with a quadruped to record terrain that wheels do not reach. It adopts the Rover's ingest contract, so every session the robots open is governed.

Figure 1. Every project on the platform is created by the same procedure and registers its data under one ingest contract. The harvesting and training agents have so far run on one of them.



Agent Workflow Diagram

Figure 2. The loop an agent runs unattended. A need arrives; the agent searches public repositories by keyword, audits what comes back, trains on what survives and scores it. The pictures that score the model also choose which epoch is reported. Review runs after every step, not only when something looks wrong, and the return edge is dashed because nothing reads the result back yet.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

agentAI is one platform for every laboratory robot

A robot, drone or sensor rig is a project with its own keys, members and access-controlled dataset registry. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

Both the Rover and the laser robot transmit data

The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial measurement units. The Laser Weed Control Robot carries two galvanometer lasers on independent channels and two cameras, one over the work zone and one forward.

Each drive becomes a dataset at its start

The platform registers a vehicle session as a governed dataset during the drive. Frames and sensor batches arrive over one API key. The laser robot transmits no position.

Every data type and domain forms its own project

On upload, a project returns an automatic diagnosis of the data before the next step is chosen. Weed detection is one project; the Rover and the laser robot are two others.

The platform is live; the QR code links to it

lab-b660m-c.tailfa6424.ts.net

Authentication requires an institutional Google account.

OPEN-WEIGHT MODELS REVIEW IN TIERS

Citing evidence beats model size

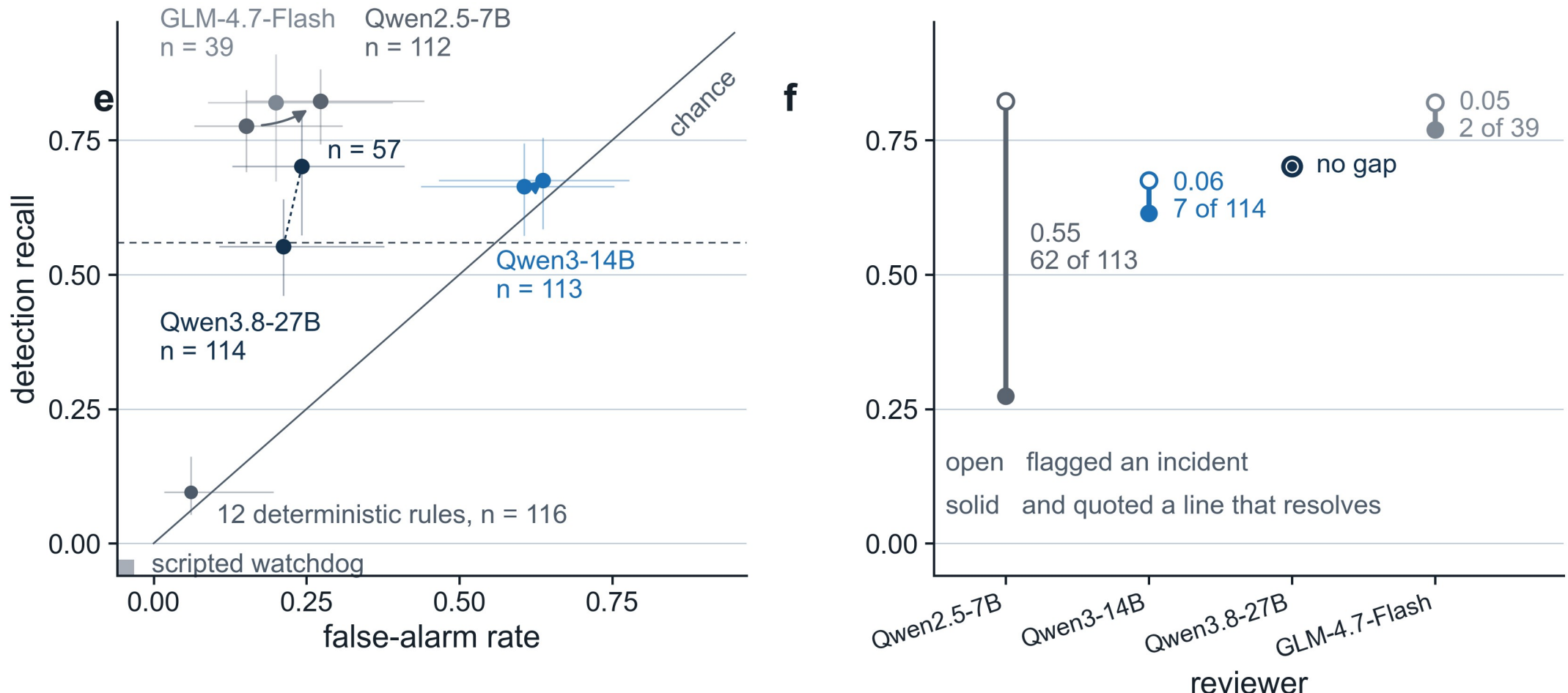


Figure 3. (e) Recall and (f) grounded recall on 149 frozen failures taken from this project's own record, scored by the project's own scorer. Each arrow runs from the model reading the raw files to the same model handed the twelve checks' findings. Grounded counts only the cases where the model also quoted a line that resolves the failure. The dotted arrow carries no head: its two ends were scored on different cases, 57 against 114.

We froze 149 real failures from our own history and asked open-weight models to find them. Good reviewers cite their evidence; size matters less. Handed the twelve checks' findings, a 27-billion model found 0.149 more, a 14-billion one 0.012.

Review runs even when nothing looks wrong

Review follows every step: failures here announced themselves as successes. Four levels by cost: the running watchdog, twelve fixed checks, a model reading files, then that model given the findings. The watchdog returned nothing on 149 frozen failures.

WHAT THE AGENTS DECIDED ON THEIR OWN

What the lab model says stays a draft

On the cluster	
plan the next step	qwen3.8 27B
review in depth	glm-4.7-flash
hard one-off reasoning	deepseek-v3 671B
decide what to collect	gemma 4
judge dataset quality	qwen2.5 7B
label assistance	minicpm-v

On the lab box

turn intent into a project	qwen2.5 3B
summarise a dataset	qwen2.5 3B

filled square: the role exercises judgement

Figure 4. The eight jobs that need a model, and the model each one was given, read from the router's own table. A filled square marks a job that exercises judgement. The six on the cluster come back authoritative; the two in the lab come back marked a draft.

Eight jobs need a model. Most run on the cluster; two run in the lab on a 3B model, because a person is waiting. A lab answer is marked so it cannot count as a decision. Nine campaign reviews predate the mark: a 4.7 GB model wrote them while 458 GB sat idle.

The model's explanation failed its own test

Downloads carry no plant name, so every box from a site gets the same invented label. A model blamed that and wrote down first what would disprove it. A human approved; four runs disproved it. Web pictures average nine boxes, ours under two.

WHAT THE AGENTS DECIDED ON THEIR OWN (CONT.)

The search ran dry after round three

Two programs ran on the cluster unwatched, one searching for data, one training on it. They completed fifteen rounds. After the third the search found nothing new and the training set never changed, though the ledger and each round report logged new work.

WHAT THE COLLECTION PRODUCED

156,521 labelled by machine, 3,208 kept

A machine labelled 156,521 pictures; 59,134 remain once duplicate downloads are removed. We audited 13,527 of them, and only 3,208 came from a source worth keeping. The checker scores 1.000 on hand-labelled pictures, so a low score means the source is poor.

Gemma 4 said yes to every box

We cropped each box out and asked Gemma 4, a vision model, whether it held a weed. It said yes every time, wrong ones included, so it cannot serve as a check. Distance from trusted examples, on DINOv2 features, does the work. Fifty-one datasets were cut.

One bad box hides from a dataset filter

A labelling mistake is usually one bad box in an otherwise sound picture, invisible to a whole-dataset filter. The check must run box by box. We built those tools; their one run died on a bookkeeping bug and was never repeated, so we cannot say they work.